

Lab Session 01

Q7:

3-Input AND Gate:

S#	X1	X2	X3	Z	W1i	W2i	W3i	K	Y	D	W1f	W2f	W3f	bias	alpha	Threshold
1	0	0	0	0	1	1	1	0	0	0	1	1	1	0	0.1	0.5
	0	0	1	0	1	1	1	1	1	-1	1	1	0.9	0	0.1	0.5
	0	1	0	0	1	1	0.9	1	1	-1	1	0.9	1	0	0.1	0.5
	0	1	1	0	1	0.9	1	1.9	1	-1	1	0.9	0.9	0	0.1	0.5
	1	0	0	0	1	0.9	0.9	1	1	-1	0.9	1	1	0	0.1	0.5
	1	0	1	0	0.9	1	1	1.9	1	-1	0.8	0.9	0.8	0	0.1	0.5
	1	1	0	0	0.8	0.9	0.8	1.7	1	-1	0.7	0.7	0.8	0	0.1	0.5
	1	1	1	1	0.7	0.7	0.8	2.2	1	0	0.7	0.7	0.7	0	0.1	0.5
2	0	0	0	0	0.7	0.7	0.7	0	0	0	0.7	0.7	0.7	0	0.1	0.5
	0	0	1	0	0.7	0.7	0.7	0.7	1	-1	0.7	0.7	0.6	0	0.1	0.5
	0	1	0	0	0.7	0.7	0.6	0.7	1	-1	0.7	0.6	0.7	0	0.1	0.5
	0	1	1	0	0.7	0.6	0.7	1.3	1	-1	0.7	0.6	0.6	0	0.1	0.5
	1	0	0	0	0.7	0.6	0.6	0.7	1	-1	0.6	0.7	0.7	0	0.1	0.5
	1	0	1	0	0.6	0.7	0.7	1.3	1	-1	0.5	0.6	0.5	0	0.1	0.5
	1	1	0	0	0.5	0.6	0.5	1.1	1	-1	0.4	0.4	0.5	0	0.1	0.5
	1	1	1	1	0.4	0.4	0.5	1.3	1	0	0.4	0.4	0.4	0	0.1	0.5
3	0	0	0	0	0.4	0.4	0.4	0	0	0	0.4	0.4	0.4	0	0.1	0.5
	0	0	1	0	0.4	0.4	0.4	0.4	0	0	0.4	0.4	0.4	0	0.1	0.5
	0	1	0	0	0.4	0.4	0.4	0.4	0	0	0.4	0.4	0.4	0	0.1	0.5
	0	1	1	0	0.4	0.4	0.4	0.8	1	-1	0.4	0.3	0.3	0	0.1	0.5
	1	0	0	0	0.4	0.3	0.3	0.4	0	0	0.4	0.4	0.4	0	0.1	0.5
	1	0	1	0	0.4	0.4	0.4	0.8	1	-1	0.3	0.4	0.3	0	0.1	0.5
	1	1	0	0	0.3	0.4	0.3	0.7	1	-1	0.2	0.2	0.3	0	0.1	0.5
	1	1	1	1	0.2	0.2	0.3	0.7	1	0	0.2	0.2	0.2	0	0.1	0.5
4	0	0	0	0	0.2	0.2	0.2	0	0	0	0.2	0.2	0.2	0	0.1	0.5
	0	0	1	0	0.2	0.2	0.2	0.2	0	0	0.2	0.2	0.2	0	0.1	0.5
	0	1	0	0	0.2	0.2	0.2	0.2	0	0	0.2	0.2	0.2	0	0.1	0.5
	0	1	1	0	0.2	0.2	0.2	0.4	0	0	0.2	0.2	0.2	0	0.1	0.5
	1	0	0	0	0.2	0.2	0.2	0.2	0	0	0.2	0.2	0.2	0	0.1	0.5
	1	0	1	0	0.2	0.2	0.2	0.4	0	0	0.2	0.2	0.2	0	0.1	0.5
	1	1	0	0	0.2	0.2	0.2	0.4	0	0	0.2	0.2	0.2	0	0.1	0.5
	1	1	1	1	0.2	0.2	0.2	0.6	1	0	0.2	0.2	0.2	0	0.1	0.5

3-Input OR Gate:

S#	X1	X2	X3	Z	W1i	W2i	W3i	K	Y	D	W1f	W2f	W3f	bias	alpha	Threshold
1	0	0	0	0	1	1	1	0	0	0	1	1	1	0	0.1	0.5
	0	0	1	1	1	1	1	1	1	0	1	1	1	0	0.1	0.5
	0	1	0	1	1	1	1	1	1	0	1	1	1	0	0.1	0.5
	0	1	1	1	1	1	1	2	1	0	1	1	1	0	0.1	0.5
	1	0	0	1	1	1	1	1	1	0	1	1	1	0	0.1	0.5
	1	0	1	1	1	1	1	2	1	0	1	1	1	0	0.1	0.5
	1	1	0	1	1	1	1	2	1	0	1	1	1	0	0.1	0.5
	1	1	1	1	1	1	1	3	1	0	1	1	1	0	0.1	0.5
2	0	0	0	0	1	1	1	0	0	0	1	1	1	0	0.1	0.5
	0	0	1	1	1	1	1	1	1	0	1	1	1	0	0.1	0.5
	0	1	0	1	1	1	1	1	1	0	1	1	1	0	0.1	0.5
	0	1	1	1	1	1	1	2	1	0	1	1	1	0	0.1	0.5
	1	0	0	1	1	1	1	1	1	0	1	1	1	0	0.1	0.5
	1	0	1	1	1	1	1	2	1	0	1	1	1	0	0.1	0.5
	1	1	0	1	1	1	1	2	1	0	1	1	1	0	0.1	0.5
	1	1	1	1	1	1	1	3	1	0	1	1	1	0	0.1	0.5

3-Input NAND Gate:

S#	X1	X2	X3	Z	W1i	W2i	W3i	K	Y	D	W1f	W2f	W3f	bias	alpha	Threshold
1	0	0	0	1	0.1	0.1	0.1	1	1	0	0.1	0.1	0.1	1	0.1	0.5
	0	0	1	1	0.1	0.1	0.1	1.1	1	0	0.1	0.1	0.1	1	0.1	0.5
	0	1	0	1	0.1	0.1	0.1	1.1	1	0	0.1	0.1	0.1	1	0.1	0.5
	0	1	1	1	0.1	0.1	0.1	1.2	1	0	0.1	0.1	0.1	1	0.1	0.5
	1	0	0	1	0.1	0.1	0.1	1.1	1	0	0.1	0.1	0.1	1	0.1	0.5
	1	0	1	1	0.1	0.1	0.1	1.2	1	0	0.1	0.1	0.1	1	0.1	0.5
	1	1	0	1	0.1	0.1	0.1	1.2	1	0	0.1	0.1	0.1	1	0.1	0.5
	1	1	1	0	0.1	0.1	0.1	1.3	1	-1	0	0	0	1	0.1	0.5
2	0	0	0	1	0	0	0	1	1	0	0	0	0	1	0.1	0.5
	0	0	1	1	0	0	0	1	1	0	0	0	0	1	0.1	0.5
	0	1	0	1	0	0	0	1	1	0	0	0	0	1	0.1	0.5
	0	1	1	1	0	0	0	1	1	0	0	0	0	1	0.1	0.5
	1	0	0	1	0	0	0	1	1	0	0	0	0	1	0.1	0.5
	1	0	1	1	0	0	0	1	1	0	0	0	0	1	0.1	0.5
	1	1	0	1	0	0	0	1	1	0	0	0	0	1	0.1	0.5
	1	1	1	0	0	0	0	1	1	-1	-0.1	-0.1	-0.1	1	0.1	0.5
3	0	0	0	1	-0.1	-0.1	-0.1	1	1	0	-0.1	-0.1	-0.1	1	0.1	0.5
	0	0	1	1	-0.1	-0.1	-0.1	0.9	1	0	-0.1	-0.1	-0.1	1	0.1	0.5
	0	1	0	1	-0.1	-0.1	-0.1	0.9	1	0	-0.1	-0.1	-0.1	1	0.1	0.5
	0	1	1	1	-0.1	-0.1	-0.1	0.8	1	0	-0.1	-0.1	-0.1	1	0.1	0.5
	1	0	0	1	-0.1	-0.1	-0.1	0.9	1	0	-0.1	-0.1	-0.1	1	0.1	0.5
	1	0	1	1	-0.1	-0.1	-0.1	0.8	1	0	-0.1	-0.1	-0.1	1	0.1	0.5
	1	1	0	1	-0.1	-0.1	-0.1	0.8	1	0	-0.1	-0.1	-0.1	1	0.1	0.5
	1	1	1	0	-0.1	-0.1	-0.1	0.7	1	-1	-0.2	-0.2	-0.2	1	0.1	0.5
4	0	0	0	1	-0.2	-0.2	-0.2	1	1	0	-0.2	-0.2	-0.2	1	0.1	0.5
	0	0	1	1	-0.2	-0.2	-0.2	0.8	1	0	-0.2	-0.2	-0.2	1	0.1	0.5
	0	1	0	1	-0.2	-0.2	-0.2	0.8	1	0	-0.2	-0.2	-0.2	1	0.1	0.5
	0	1	1	1	-0.2	-0.2	-0.2	0.6	1	0	-0.2	-0.2	-0.2	1	0.1	0.5
	1	0	0	1	-0.2	-0.2	-0.2	0.8	1	0	-0.2	-0.2	-0.2	1	0.1	0.5
	1	0	1	1	-0.2	-0.2	-0.2	0.6	1	0	-0.2	-0.2	-0.2	1	0.1	0.5
	1	1	0	1	-0.2	-0.2	-0.2	0.6	1	0	-0.2	-0.2	-0.2	1	0.1	0.5
	1	1	1	0	-0.2	-0.2	-0.2	0.4	0	0	-0.2	-0.2	-0.2	1	0.1	0.5

3-Input NOR Gate:

S#	X1	X2	X3	Z	W1i	W2i	W3i	K	Y	D	W1f	W2f	W3f	bias	alpha	Threshold
1	0	0	0	1	0.1	0.1	0.1	1	1	0	0.1	0.1	0.1	1	0.1	0.5
	0	0	1	0	0.1	0.1	0.1	1.1	1	-1	0.1	0.1	0	1	0.1	0.5
	0	1	0	0	0.1	0.1	0	1.1	1	-1	0.1	0	0.1	1	0.1	0.5
	0	1	1	0	0.1	0	0.1	1.1	1	-1	0.1	0	0	1	0.1	0.5
	1	0	0	0	0.1	0	0	1.1	1	-1	0	0.1	0.1	1	0.1	0.5
	1	0	1	0	0	0.1	0.1	1.1	1	-1	-0.1	0	-0.1	1	0.1	0.5
	1	1	0	0	-0.1	0	-0.1	0.9	1	-1	-0.2	-0.2	-0.1	1	0.1	0.5
	1	1	1	0	-0.2	-0.2	-0.1	0.5	0	0	-0.2	-0.2	-0.2	1	0.1	0.5
2	0	0	0	1	-0.2	-0.2	-0.2	1	1	0	-0.2	-0.2	-0.2	1	0.1	0.5
	0	0	1	0	-0.2	-0.2	-0.2	0.8	1	-1	-0.2	-0.2	-0.3	1	0.1	0.5
	0	1	0	0	-0.2	-0.2	-0.3	0.8	1	-1	-0.2	-0.3	-0.2	1	0.1	0.5
	0	1	1	0	-0.2	-0.3	-0.2	0.5	0	0	-0.2	-0.2	-0.2	1	0.1	0.5
	1	0	0	0	-0.2	-0.2	-0.2	0.8	1	-1	-0.3	-0.2	-0.2	1	0.1	0.5
	1	0	1	0	-0.3	-0.2	-0.2	0.5	0	0	-0.3	-0.3	-0.3	1	0.1	0.5
	1	1	0	0	-0.3	-0.3	-0.3	0.4	0	0	-0.3	-0.3	-0.3	1	0.1	0.5
	1	1	1	0	-0.3	-0.3	-0.3	0.1	0	0	-0.3	-0.3	-0.3	1	0.1	0.5
3	0	0	0	1	-0.3	-0.3	-0.3	1	1	0	-0.3	-0.3	-0.3	1	0.1	0.5
	0	0	1	0	-0.3	-0.3	-0.3	0.7	1	-1	-0.3	-0.3	-0.4	1	0.1	0.5
	0	1	0	0	-0.3	-0.3	-0.4	0.7	1	-1	-0.3	-0.4	-0.3	1	0.1	0.5
	0	1	1	0	-0.3	-0.4	-0.3	0.3	0	0	-0.3	-0.3	-0.3	1	0.1	0.5
	1	0	0	0	-0.3	-0.3	-0.3	0.7	1	-1	-0.4	-0.3	-0.3	1	0.1	0.5
	1	0	1	0	-0.4	-0.3	-0.3	0.3	0	0	-0.4	-0.4	-0.4	1	0.1	0.5
	1	1	0	0	-0.4	-0.4	-0.4	0.2	0	0	-0.4	-0.4	-0.4	1	0.1	0.5
	1	1	1	0	-0.4	-0.4	-0.4	-0.2	0	0	-0.4	-0.4	-0.4	1	0.1	0.5
4	0	0	0	1	-0.4	-0.4	-0.4	1	1	0	-0.4	-0.4	-0.4	1	0.1	0.5
	0	0	1	0	-0.4	-0.4	-0.4	0.6	1	-1	-0.4	-0.4	-0.5	1	0.1	0.5
	0	1	0	0	-0.4	-0.4	-0.5	0.6	1	-1	-0.4	-0.5	-0.4	1	0.1	0.5
	0	1	1	0	-0.4	-0.5	-0.4	0.1	0	0	-0.4	-0.4	-0.4	1	0.1	0.5
	1	0	0	0	-0.4	-0.4	-0.4	0.6	1	-1	-0.5	-0.4	-0.4	1	0.1	0.5
	1	0	1	0	-0.5	-0.4	-0.4	0.1	0	0	-0.5	-0.5	-0.5	1	0.1	0.5
	1	1	0	0	-0.5	-0.5	-0.5	0	0	0	-0.5	-0.5	-0.5	1	0.1	0.5
	1	1	1	0	-0.5	-0.5	-0.5	-0.5	0	0	-0.5	-0.5	-0.5	1	0.1	0.5
5	0	0	0	1	-0.5	-0.5	-0.5	1	1	0	-0.5	-0.5	-0.5	1	0.1	0.5
	0	0	1	0	-0.5	-0.5	-0.5	0.5	0	0	-0.5	-0.5	-0.5	1	0.1	0.5
	0	1	0	0	-0.5	-0.5	-0.5	0.5	0	0	-0.5	-0.5	-0.5	1	0.1	0.5
	0	1	1	0	-0.5	-0.5	-0.5	0	0	0	-0.5	-0.5	-0.5	1	0.1	0.5
	1	0	0	0	-0.5	-0.5	-0.5	0.5	0	0	-0.5	-0.5	-0.5	1	0.1	0.5
	1	0	1	0	-0.5	-0.5	-0.5	0	0	0	-0.5	-0.5	-0.5	1	0.1	0.5
	1	1	0	0	-0.5	-0.5	-0.5	0	0	0	-0.5	-0.5	-0.5	1	0.1	0.5
	1	1	1	0	-0.5	-0.5	-0.5	-0.5	0	0	-0.5	-0.5	-0.5	1	0.1	0.5

2-Input XOR Training:

OR Gate Implementation													
S#	X1	X2	Z	W1i	W2i	K	Y	D	W1f	W2f	bias	alpha	Threshold
1	0	0	0	0.1	0.1	0	0	0	0.1	0.1	0	0.1	0.5
	0	1	1	0.1	0.1	0.1	0	1	0.1	0.2	0	0.1	0.5
	1	0	1	0.1	0.2	0.1	0	1	0.2	0.2	0	0.1	0.5
	1	1	1	0.2	0.2	0.4	0	1	0.3	0.3	0	0.1	0.5
2	0	0	0	0.3	0.3	0	0	0	0.3	0.3	0	0.1	0.5
	0	1	1	0.3	0.3	0.3	0	1	0.3	0.4	0	0.1	0.5
	1	0	1	0.3	0.4	0.3	0	1	0.4	0.4	0	0.1	0.5
	1	1	1	0.4	0.4	0.8	1	0	0.4	0.4	0	0.1	0.5
3	0	0	0	0.4	0.4	0	0	0	0.4	0.4	0	0.1	0.5
	0	1	1	0.4	0.4	0.4	0	1	0.4	0.5	0	0.1	0.5
	1	0	1	0.4	0.5	0.4	0	1	0.5	0.5	0	0.1	0.5
	1	1	1	0.5	0.5	1	1	0	0.5	0.5	0	0.1	0.5
4	0	0	0	0.5	0.5	0	0	0	0.5	0.5	0	0.1	0.5
	0	1	1	0.5	0.5	0.5	0	1	0.5	0.6	0	0.1	0.5
	1	0	1	0.5	0.6	0.5	0	1	0.6	0.6	0	0.1	0.5
	1	1	1	0.6	0.6	1.2	1	0	0.6	0.6	0	0.1	0.5
5	0	0	0	0.6	0.6	0	0	0	0.6	0.6	0	0.1	0.5
	0	1	1	0.6	0.6	0.6	1	0	0.6	0.6	0	0.1	0.5
	1	0	1	0.6	0.6	0.6	1	0	0.6	0.6	0	0.1	0.5
	1	1	1	0.6	0.6	1.2	1	0	0.6	0.6	0	0.1	0.5

NAND Gate Implementation													
S#	X1	X2	Z	W1i	W2i	K	Y	D	W1f	W2f	bias	alpha	Threshold
1	0	0	1	0.1	0.1	1	1	0	0.1	0.1	1	0.1	0.5
	0	1	1	0.1	0.1	1.1	1	0	0.1	0.1	1	0.1	0.5
	1	0	1	0.1	0.1	1.1	1	0	0.1	0.1	1	0.1	0.5
	1	1	0	0.1	0.1	1.2	1	-1	0	0	1	0.1	0.5
2	0	0	1	0	0	1	1	0	0	0	1	0.1	0.5
	0	1	1	0	0	1	1	0	0	0	1	0.1	0.5
	1	0	1	0	0	1	1	0	0	0	1	0.1	0.5
	1	1	0	0	0	1	1	-1	-0.1	-0.1	1	0.1	0.5
3	0	0	1	-0.1	-0.1	1	1	0	-0.1	-0.1	1	0.1	0.5
	0	1	1	-0.1	-0.1	0.9	1	0	-0.1	-0.1	1	0.1	0.5
	1	0	1	-0.1	-0.1	0.9	1	0	-0.1	-0.1	1	0.1	0.5
	1	1	0	-0.1	-0.1	0.8	1	-1	-0.2	-0.2	1	0.1	0.5
4	0	0	1	-0.2	-0.2	1	1	0	-0.2	-0.2	1	0.1	0.5
	0	1	1	-0.2	-0.2	0.8	1	0	-0.2	-0.2	1	0.1	0.5
	1	0	1	-0.2	-0.2	0.8	1	0	-0.2	-0.2	1	0.1	0.5
	1	1	0	-0.2	-0.2	0.6	1	-1	-0.3	-0.3	1	0.1	0.5
5	0	0	1	-0.3	-0.3	1	1	0	-0.3	-0.3	1	0.1	0.5
	0	1	1	-0.3	-0.3	0.7	1	0	-0.3	-0.3	1	0.1	0.5
	1	0	1	-0.3	-0.3	0.7	1	0	-0.3	-0.3	1	0.1	0.5
	1	1	0	-0.3	-0.3	0.4	0	0	-0.3	-0.3	1	0.1	0.5

AND Gate Implementation (XOR Gate Formed)													
S#	X1	X2	Z	W1i	W2i	K	Y	D	W1f	W2f	bias	alpha	Threshold
1	0	1	0	0.1	0.1	0.1	0	0	0.1	0.1	0	0.1	0.5
	1	1	1	0.1	0.1	0.2	0	1	0.2	0.2	0	0.1	0.5
	1	1	1	0.2	0.2	0.4	0	1	0.3	0.3	0	0.1	0.5
	1	0	0	0.3	0.3	0.3	0	0	0.3	0.3	0	0.1	0.5
2	0	0	0	0.3	0.3	0	0	0	0.3	0.3	0	0.1	0.5
	1	1	1	0.3	0.3	0.6	1	0	0.3	0.3	0	0.1	0.5
	1	1	1	0.3	0.3	0.6	1	0	0.3	0.3	0	0.1	0.5
	1	0	0	0.3	0.3	0.3	0	0	0.3	0.3	0	0.1	0.5

2-Input XNOR Training:

AND Gate Implementation													
S#	X1	X2	Z	W1i	W2i	K	Y	D	W1f	W2f	bias	alpha	Threshold
1	0	0	0	0.1	0.1	0	0	0	0.1	0.1	0	0.1	0.5
	0	1	0	0.1	0.1	0.1	0	0	0.1	0.1	0	0.1	0.5
	1	0	0	0.1	0.1	0.1	0	0	0.1	0.1	0	0.1	0.5
	1	1	1	0.1	0.1	0.2	0	1	0.2	0.2	0	0.1	0.5
2	0	0	0	0.2	0.2	0	0	0	0.2	0.2	0	0.1	0.5
	0	1	0	0.2	0.2	0.2	0	0	0.2	0.2	0	0.1	0.5
	1	0	0	0.2	0.2	0.2	0	0	0.2	0.2	0	0.1	0.5
	1	1	1	0.2	0.2	0.4	0	1	0.3	0.3	0	0.1	0.5
3	0	0	0	0.3	0.3	0	0	0	0.3	0.3	0	0.1	0.5
	0	1	0	0.3	0.3	0.3	0	0	0.3	0.3	0	0.1	0.5
	1	0	0	0.3	0.3	0.3	0	0	0.3	0.3	0	0.1	0.5
	1	1	1	0.3	0.3	0.6	1	0	0.3	0.3	0	0.1	0.5
4	0	0	0	0.3	0.3	0	0	0	0.3	0.3	0	0.1	0.5
	0	1	0	0.3	0.3	0.3	0	0	0.3	0.3	0	0.1	0.5
	1	0	0	0.3	0.3	0.3	0	0	0.3	0.3	0	0.1	0.5
	1	1	1	0.3	0.3	0.6	1	0	0.3	0.3	0	0.1	0.5

NOR Gate Implementation													
S#	X1	X2	Z	W1i	W2i	K	Y	D	W1f	W2f	bias	alpha	Threshold
1	0	0	1	0.1	0.1	1	1	0	0.1	0.1	1	0.1	0.5
	0	1	0	0.1	0.1	1.1	1	-1	0.1	0	1	0.1	0.5
	1	0	0	0.1	0	1.1	1	-1	0	0	1	0.1	0.5
	1	1	0	0	0	1	1	-1	-0.1	-0.1	1	0.1	0.5
2	0	0	1	-0.1	-0.1	1	1	0	-0.1	-0.1	1	0.1	0.5
	0	1	0	-0.1	-0.1	0.9	1	-1	-0.1	-0.2	1	0.1	0.5
	1	0	0	-0.1	-0.2	0.9	1	-1	-0.2	-0.2	1	0.1	0.5
	1	1	0	-0.2	-0.2	0.6	1	-1	-0.3	-0.3	1	0.1	0.5
3	0	0	1	-0.3	-0.3	1	1	0	-0.3	-0.3	1	0.1	0.5
	0	1	0	-0.3	-0.3	0.7	1	-1	-0.3	-0.4	1	0.1	0.5
	1	0	0	-0.3	-0.4	0.7	1	-1	-0.4	-0.4	1	0.1	0.5
	1	1	0	-0.4	-0.4	0.2	0	0	-0.4	-0.4	1	0.1	0.5
4	0	0	1	-0.4	-0.4	1	1	0	-0.4	-0.4	1	0.1	0.5
	0	1	0	-0.4	-0.4	0.6	1	-1	-0.4	-0.5	1	0.1	0.5
	1	0	0	-0.4	-0.5	0.6	1	-1	-0.5	-0.5	1	0.1	0.5
	1	1	0	-0.5	-0.5	0	0	0	-0.5	-0.5	1	0.1	0.5
5	0	0	1	-0.5	-0.5	1	1	0	-0.5	-0.5	1	0.1	0.5
	0	1	0	-0.5	-0.5	0.5	0	0	-0.5	-0.5	1	0.1	0.5
	1	0	0	-0.5	-0.5	0.5	0	0	-0.5	-0.5	1	0.1	0.5
	1	1	0	-0.5	-0.5	0	0	0	-0.5	-0.5	1	0.1	0.5

OR Gate Implementation (XNOR Gate Formed)													
S#	X1	X2	Z	W1i	W2i	K	Y	D	W1f	W2f	bias	alpha	Threshold
1	0	1	1	0.1	0.1	0.1	0	1	0.1	0.2	0	0.1	0.5
	0	0	0	0.1	0.2	0	0	0	0.1	0.2	0	0.1	0.5
	0	0	0	0.1	0.2	0	0	0	0.1	0.2	0	0.1	0.5
	1	0	1	0.1	0.2	0.1	0	1	0.2	0.2	0	0.1	0.5